Class XII Session 2025-26
Subject - Chemistry
Sample Question Paper - 5

Time Allowed: 30 Minutes

General Instructions:

Read the following instructions carefully.

1. There are 16 questions in this question paper with internal choice.
2. SECTION A consists of 16 multiple-choice questions carrying 1 mark each.
3. **All questions are compulsory.**
4. **Use of log tables and calculators is not allowed.**

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Section A

1. Clemmensen reduction of a ketone is carried out in the presence of which of the following? **[1]**

   a) $LiAlH_4$
   b) Zn – Hg with HCl
   c) H_2 and Pt as catalyst
   d) Glycol with KOH

2. When a haloalkane with β – hydrogen atom is heated with alcoholic solution of potassium hydroxide then: **[1]**

   a) elimination of hydrogen atom from β– carbon
   b) All of these
   c) alkene is formed as a product
   d) elimination of halogen atom from α – carbon

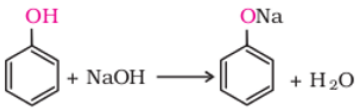
3. Which one is not the essential amino acid in the ones given below? **[1]**

   a) Proline
   b) Valine
   c) Leucine
   d) Arginine

4. The reagent which can be used to distinguish acetophenone from benzophenone is: **[1]**

   a) 2, 4-dinitrophenyl hydrazine
   b) Benedict solution
   c) aqueous solution of $NaHSO_3$
   d) I_2 and Na_2CO_3

5. In the reaction **[1]**



Sodium phenoxide

a) Cleavage of O - H bond

b) Phenols are acidic in nature.

c) All of these

d) They can donate a proton to a stronger base

6. Match the items given in column I with that in column II: **[1]**

Column I	Column II
(a) Urea	(i) i < 1
(b) $FeCl_3$	(ii) i = 1
(c) Benzoic acid in Benzene	(iii) i = 2
(d) $MgSO_4$	(iv) i = 4

a) (a) - (ii), (b) - (i), (c) - (iii), (d) - (iv)

b) (a) - (iv), (b) - (ii), (c) - (iii), (d) - (i)

c) (a) - (i), (b) - (ii), (c) - (iii), (d) - (iv)

d) (a) - (ii), (b) - (iv), (c) - (i), (d) - (iii)

7. The rate law for a particular reaction is given as rate = $k[A][B]^2$. **[1]**

How is the rate of reaction affected if we double the concentration of B?

a) becomes half ($\frac{1}{2}$)

b) four times

c) three times

d) two times

8. Ethylidene chloride is a/an _____. **[1]**

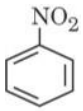
a) gem-dihalide

b) vic-dihalide

c) vinylic halide

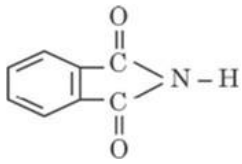
d) allylic halide

9. Hoffmann Bromamide Degradation reaction is given by: **[1]**

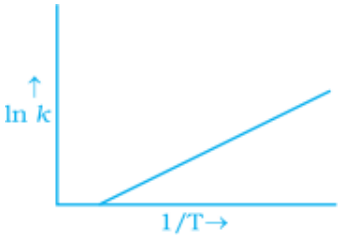
a) 

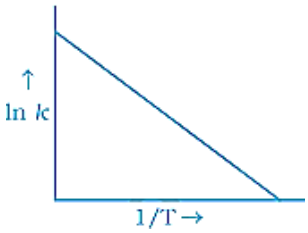
b) $CH_3 - CH_2 - \overset{O}{\overset{||}{C}} - NH_2$

c) $CH_3 - C \equiv N$

d) 

10. According to the Arrhenius equation rate constant k is equal to $Ae^{-E_a/RT}$. Which of the following options represents the graph of ln k vs $\frac{1}{T}$? **[1]**

a) 

b) 

c)

d)

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11. One of the following alcohols do not undergo oxidation reaction: **[1]**

a) Tertiary alcohol
b) Primary alcohol and Secondary alcohol
c) Secondary alcohol
d) Primary alcohol

12. The incorrect statement about interstitial compounds is: **[1]**

a) They have high melting point.
b) They retain metallic conductivity.
c) They are chemically reactive.
d) They are very hard.

13. **Assertion (A):** D (+)-Glucose is dextrorotatory. **[1]**
Reason (R): Symbol 'D' represents its dextrorotatory nature.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

14. **Assertion (A):** In comparison to ethyl chloride it is difficult to carry out nucleophilic substitution on vinyl chloride. **[1]**
Reason (R): Vinyl group is electron-donating.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

15. **Assertion (A):** Phenol gives 2,4,6-trinitrophenol on treatment with concentrated HNO_3 and concentrated H_2SO_4. **[1]**
Reason (R): -OH group in phenol is m-directing.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

16. **Assertion (A):** RMgX on reaction with CO_2 gives $R-\overset{\overset{O}{||}}{C}-OH$ **[1]**
Reason (R): RMgX on reaction with carbonyl compounds give alcohols.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.

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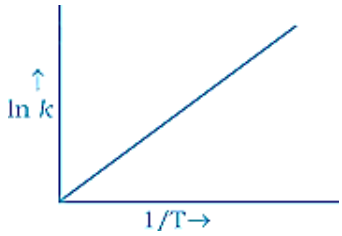
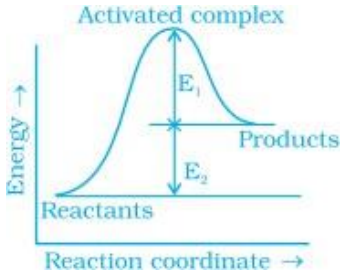
c) A is true but R is false.



d) A is false but R is true.



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